

Blockchain Technology in the Public Sector by Leveraging the Triumvirate of Security

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Abstract—Blockchain technology has become a trend in various sectors, both industrial and public sectors. There have been many questions regarding this blockchain technology in terms of competition in the public sector. Mainly among these are the concerns that become issues around governance and control in distributed ledgers. From an information security point of view this article investigates the influence of blockchain technology on public domain processes. It contains an outline of the development of the provision of public services and one-way information by the government to the community, the exchange of information and communication between the public and the government through the current blockchain application, and the current development of blockchain technology. We use a triad, Confidentiality-Integrity-Accessibility to guide our discussion of the interrelationships of security, operation, and regulation of this generation. By leveraging the triad model, we offer a framework for public managers who may also be able to recommend blockchain technology, and we observe the positive benefits that arise from the denial of a distributed ledger. In particular, we look at the benefits of blockchain technology which has to do with disclaimers to help public managers understand how to leverage blockchain generation for an application process.

Keywords—Blockchain, Public Sector, Public Service, Government, CIA

I. INTRODUCTION

Cyber assaults keep threatening the safety of statistics belonging to authority's agencies in Indonesia. Currently, maximum human beings have switched to the usage of virtual structures to perform numerous activities. Advances in the statistics era that exist presently are very influential on human lifestyles today, due to the fact the improvement of era can assist and facilitate a job, in order that it is able to make powerful decisions [1]. Especially during the cutting-edge pandemic, human beings are recommended to live at home in order that they use virtual structures extra to do numerous things, be it for online transactions, studying, running and others [2]. On the opposite hand, cyber hackers keep taking

advantage of the instant to devote a chain of cybercrimes. This isn't any exception during a virus like today [3].

In 2020 alone, the National Cyber and Crypto Agency has discovered that there was a 4-fold boom in cyber assaults. Cyber assaults that arise can take the shape of malware, trojans, phishing, and others. Therefore, during this pandemic, BSSN urges enterprise proprietors to keep enhancing statistics protection regulations to assume cyber assaults [4]. By making top arrangements for statistics protection, authorities' agencies can reduce the incidence of statistics robbery or cyber incidents.

Blockchain technology, which first evolved in 2008, could change the variety of offerings provided through any authority organization at any level [5]. So, what exactly is blockchain technology? A distributed ledger, often referred to as a blockchain, updates a primary database (e.g., transaction data or identities), using a peer-to-peer decentralized community at the top of the Internet to create a shared virtual ledger that continues immutable documents of every transaction. It's an open, allocated, communal device, obviously without any vulnerability factors [6]. Since all participants (e.g., all events in the delivery chain, economic transactions, commercial enterprise processes, or authority projects) have access rights to the information, it cannot be changed [7]. Companies will develop information security policies to handle and protect important information or data in their possession. The policies that have been developed will apply to all IT structures [8].

In addition, information security policies are also useful for determining who has access to various types of data, how identities will be authenticated, what methods are used to secure information, and so on. These steps will help you prevent the risk of theft, modification, or loss of important information [9].

Please note, that most information security policies will have a focus or goal that refers to the CIA triad aspect. These aspects are confidentiality, integrity, and availability. Any attack in cyberspace will generally try to violate at least one aspect or attribute of the CIA triad [10]. Having a good understanding of this information security model will help you better minimize the risk of attacks or system errors. With

the right security policies, you can protect important information from unauthorized activities [11].

Anxiety about security became one of the reasons for taking up blockchain technology. Recently published factors revealed the lack of systematic opinion literature focusing on blockchain applications on internet security issues [12]. As large data stores become individually identifiable residents, and more providers of mobile services and transactions, government officials are facing increasing problems in ensuring reliable and secure solutions [13]. Famous for the creation of the cryptocurrency Bitcoin, Blockchain offers a very extraordinary settlement that is influential for the era of innovation [14]. As we remember the security associated with blockchain, scattered data can be spared from being altered. The government is still exploring opportunities to use blockchain technology as a form of service benefit beyond financial transactions, government-to-citizen interactions, government-to-government, and government-to-vendor services throughout [15].

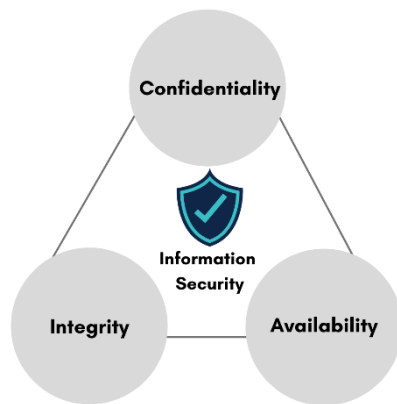


Figure 1. The Triumvirate of Security

In line with the rapidly increasing use of blockchain and the need to investigate issues related to potentially transformative technologies, this article examines the impact of blockchain on public sector procedures through reference to information security [16]. Specifically, this article uses the CIA triad to guide the opinion of 3 key areas, where blockchain can make tremendous changes to government services: security, governance, and regulation to guide citizens.

II. METHOD

A researcher needs to work at the stage of conducting research using the research method utilized by the author to attain a goal of acquiring correct and reliable information. To solve the author's difficulties and reach the research method's aims, the author employed the literature review research technique from earlier research that is related to the title [17]. In this research, there are 7 (seven) literature studies that serve as research objects and serve as benchmarks for comparison of ongoing research.

Maupin et al explain that as in early e-government focused research, early work on Blockchain was both descriptive and prescriptive, focusing on blockchain primers that are developing as disruptive technologies with a diverse portfolio possibility [18]. through leveraging secure, dispersed, and

cost-effective technology, governments will be able to innovate [19].

Company, et al [20] said closely related to the development and widespread implementation of an exempt regulatory environment, expressly for the development and adoption of blockchain solutions, is referred to as governance. Most of the new agreements are focused on the human side of finance, with the initial goal of reducing the worldwide risk that cryptocurrencies bring to government and financial institutions. Furthermore, much like early attempts to regulate the Internet, the discussion about blockchain-focused governance frameworks focuses mostly on self-regulation, which does not significantly impede innovation.

Zaharchuk [21] explained that leveraging blockchain to reengineer the net control of conventional citizen offerings may want to drastically enhance the authentication of man or woman citizens' identities and transaction facts located in economic transactions, voting, tax collection, registration, licensing, and different recurring authority's movements that require identification authentication. for validity. Second, blockchain can successfully facilitate steady audit trails for regulatory compliance to lessen agreement control expenses and enhance enforcement [22]. Third, numerous authority's agencies, specifically on the federal level, are investigating blockchain for the established order of steady and dependable identification control, specifically for immigrants and overseas visitors. Blockchain generation can enhance authorities' agreement control processes by validating completion of achievement or compliance with deadlines, thereby increasing contractor confidence.

Janssen et al [23] explained that e-government research, still applying the adoption and trust model, can leap forward with blockchain technology, which facilitates peer-to-peer transactions without intermediaries. And gaining attention among academics and practitioners alike, blockchain can influence adoption and trust models.

Alketbi et al [24] point to the success of initial government initiatives in Estonia and Australia and provide guidelines for others. Overall, blockchain generation can be utilized by organizations and governments around the arena to enhance overall performance with the aid of supplying stable and dependable documented information of information and metadata, throughout sports and processes.

Afzori [25] in the paper describes the uniqueness of blockchain as its potential to give people and groups the cappotential to remodel their interactions in politics, business, and society at huge, with an extraordinary procedure of disintermediation on a huge scale, primarily based totally on computerized and trustless transactions.

Treiblmaier et al [26] said the introduction of the blockchain era has created excessive expectations of approximately transformation throughout enterprise and government. The writer has formerly been disruptive in enterprise sectors as varied as tourism, energy, finance, and public governance. Simultaneously, numerous authors warn in opposition to unrealistic expectancies and advise cautious scrutiny of every capability use case to become aware of how blockchain era sincerely solves pending issues, implying reputation of the restrictions of blockchain and the brand-new demanding situations provided and advocating open documentation. from any restricting effect.

From the 7 (seven) reviews above, it explains the use of the blockchain era that may be implemented in numerous sectors. Blockchain era itself is a digital-primarily based totally recording gadget and can't be modified with the aid of anyone. It makes its very own alternatives for the blockchain and its users [27]. Therefore, why the blockchain era is critical withinside the global government (public sector). Blockchain indicates that decentralized consensus will offer an amazing distribution of records that may cowl all activities [28].

III. RESULTS AND DISCUSSION

A. Case Study Description

The goal of managing information security is to ensure the confidentiality, integrity, and availability of valuable information assets that may be strategic, protected, sensitive, or proprietary. Illustrated in Fig. 1, it is often called the "CIA Triumvirate." The CIA triad is a standard model in information security designed to regulate and evaluate how an organization or company is when data is stored, transmitted, or processed. Every aspect of the CIA triad (Confidentiality – Integrity – Availability) will be an important component of information security [29].

When we communicate approximately the confidentiality element or the confidentiality of records, we're speaking about a chain of safeguards in order that records aren't accessed with the aid of unauthorized parties [30]. Confidential records are taken into consideration as precious records with the aid of using cyber hackers. The records focused is generally withinside the shape of purchaser records, worker records, highbrow property, or records concerning alternate secrets. That's why cyber hackers hold to search for vulnerabilities withinside the device to get entry to this critical info [31].

In general, personal records can fall into the wrong hands because of a records breach or insider threats. Several forms of assaults are generally used to get entry to personal records consisting of [32]: Man, withinside the Middle attack, Encryption breach. To shield it, there are some steps that may be used consisting of imposing two-component authentication, the use of sturdy passwords, encryption, and others [33].

In data security, integrity refers to a way or steps to preserve statistics or data from being manipulated, modified, or edited through unauthorized parties. These measures ensure the accuracy and completeness of data, as with the safety of personal data, integrity safety is likewise necessary. The third factor of the CIA triad is availability. That is, withinside the context of data security, efforts to preserve a machine usable are essential matters that want to be done. By presenting availability safety, you need to be capable of offering guarantees that structures and statistics may be accessed through authenticated customers every time the data is needed [34].

Information property may also consist of records, statistics, hardware, software programs, or different statistics resources. Others amplify this version of statistics safety and propose that statistics safety includes 4 middle characteristics: confidentiality, integrity, availability, and accountability [35]. Confidentiality is involved with protective records from unauthorized access, integrity issues protecting the validity of records in opposition to undesirable

changes, and availability refers to the accessibility of statistics (and structures) to legal people and techniques withinside the shape and layout required. Accountability refers to the cappable to hint sports done to a specific character or technique, that is undeniable. Nonrepudiation is an assurance that a person cannot deny something just like the authenticity in their signature or the act of sending a message. Every movement done on records or structures, whether done with the aid of using code or with the aid of using humans, is logged, and related to a selected person, software program function, or different "knowable" movement [36]. To discover the phenomena mentioned in this article, we compare the latest enjoyment of one of the pioneering public companies in search of expanding a public blockchain-primarily based total technique that could enhance machine records safety. After locating numerous key informants who're presently concerned in blockchain tasks in Indonesia. Because governments in any respect degrees are concerned in lots of statistics-extensive sports among establishments and residents or different companies (in addition to internally among diverse authority's entities), and due to the fact, those transactions should be recorded and maintained in dependable and correct forms, it's far critical that IT managers in any respect degrees authority's companies put in force steady structures and techniques. For this reason, authority's managers are beginning to put in force (or compare) blockchain to take advantage of its precise technological capabilities [37].

B. Blockchain System as Information Security

Confidentiality is the number one purpose of facts protection management, however blockchains aren't designed to guide or hold facts confidentiality. In fact, blockchain is designed to be transparent; Distributed database is seen to all participants. This has been addressed in some ways, however, remains a great implementation difficulty for the era in sure contexts, such as authorities. Some answers have layers of protection constructed to offer confidentiality; however, those additionally violate several of the precept blockchain guidelines and enhance their value. Government managers thinking about blockchain answers must pay unique interest to the general protection in their structures to make sure facts are confidential [38]. Integrity, the defining characteristic of a blockchain, may be volatile if the blockchain is situation to change, so it's far vital for authority's managers to evaluate blockchain providers cautiously to make sure that they best provide the capability of the original, immutable blockchain. To make sure availability, blockchain structures are platform impartial and era agnostic. While IBM and different providers provide customers a variety of proprietary blockchain answers, the underlying blockchain era is offered from any device or era environment [39].

C. Integrity in System with Blockchain

Finally, it's miles an increasing number of essentials that everyone transactions inside authorities and among authority's entities and outdoor events are recorded, documented, recorded, and archived with an excessive degree of fact integrity. Blockchain is valuable for this purpose because it presents a robust denial property. It isn't trivial for hackers or different malicious actors to tamper with their

values and properties (which include figuring out facts) in any block or blockchain because of the digital vote casting mechanism which guarantees that any discrepancy among one detail and the relaxation of the device is corrected via way of means of the values of the bulk of the crowd. The affordability of this blockchain procedure presents a dependable supply of statistics and guarantees an excessive degree of self-assurance withinside the integrity of the statistics withinside the device [40]. So, for example, in a contracting procedure while the authorities (public sector) manage a huge creation assignment or army acquisition of a couple of carriers, there are numerous transactions among and amongst many groups that should arise reliably and regularly sequentially. A disbursed ledger the usage of blockchain is right for this purpose. Consider that everyone gamers in a complicated set of transactions (all constructing contractors and subcontractors, for example, who should be paid simplest after finishing proven vetted work) should get right of entry to the identical set of transaction statistics, and there can be incentives for statistics changes. The use of blockchain in this context might make such statistics a near-not possible non-trivial practice, consequently making sure correct reasssets and trustworthiness of statistics. Therefore, interest ought to be paid to the position of device tracking to make sure that the block database renovation is genuinely disbursed [41].

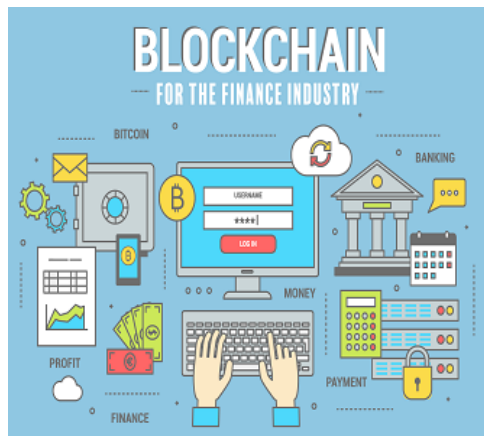


Figure 2. Application of Blockchain Technology
(Source: <https://defi-planet.com/>)

Leaders in authorized blockchain implementations have suggested that, at the same time as blockchains are public and transparent, authority companies will hold a legitimate position as the video display unit and arbiter of any conflicts that are sought to be resolved. For example, an authorized Department of Motor Vehicles may wish to administer a blockchain for the sale, license, registration, and coverage of vehicles, where multiple residents and operators participate in surveillance. In this study using the literature review method, the construction obtained is that the use of blockchain plays a very important role in the public. Because the issues that can be faced include customer protection, statistical security, stability of monetary devices, monetary integrity, cybercrime, and the effectiveness of financial policies [42].

IV. CONCLUSION

Threats for your data safety can be available in many forms. By know-how what data safety is, you could compare and become aware of employer policies. You can see if the coverage can shield private data consistent with components of the CIA triad. As defined above, withinside the CIA triad there are three primary components that need to usually be taken into consideration in keeping data safe. These components are Confidentiality, Integrity and Availability. The technological revolution promised via way of means of blockchain will permit authorities leaders to enhance system management, growth safety, and streamline transactions for efficiency, however it's going to now no longer be without sure costs. Government leaders need to apprehend the trade-offs of those rising technologies. This looks at offers a taxonomy of possibilities for implementation of blockchain packages in authority's enterprises, with a contingency technique to signify implementation details. The proposed framework will help managers as they examine the feasibility of pursuing blockchain projects. This paper makes numerous contributions. First, it makes use of a long time framework, the CIA triad, to focus on possibilities for blockchain to convert public carrier delivery. Second, combine the CIA triad with denial to discover blockchain and e-authorities. This paper additionally unites the idea for destiny studies on blockchain safety and e-authorities. Finally, it offers a name for similar studies on blockchain and authorities, regarding governance and regulatory frameworks. This challenge gives perception into a way to first-rate leverage blockchain generation to growth yields and productivity.

ACKNOWLEDGMENT

The writer may be very thankful to the Faculty of Science and Technology at the University of Raharja in Tangerang, Indonesia, for providing resources that enabled this study to be completed successfully. The writer could additionally want to thank Alphabet Incubator, University of Raharja, Ichsan Medical Centre Institute which has furnished centers and monetary aid during this research.

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